PES University — Dept. of CSE(AIML)

Lab 1: Requirements Engineering & UML Use-Case Modelling

Problem Statement #04 — Student Project Portfolio & Showcase App

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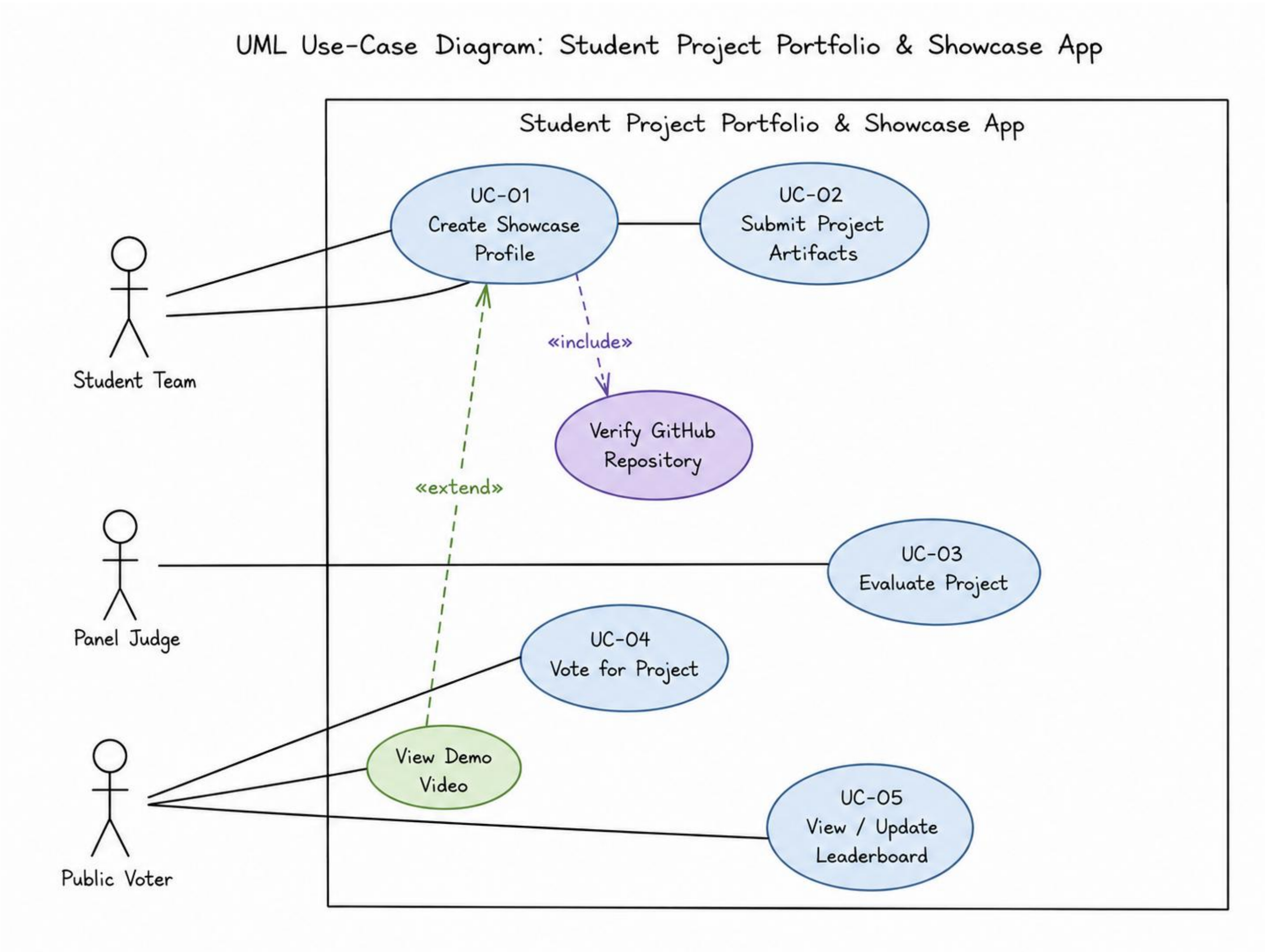
1. Requirements Table

The table below contains the requirements for the Student Project Portfolio & Showcase App. It is formatted for the combined final PDF submission.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable student teams to create showcase profiles containing abstract summaries, demo video links, and verified GitHub repository URLs.	High	Pass: Profile is published upon successful validation of the GitHub repository URL. Fail: An invalid repository URL is accepted.	Core functionality for presenting student projects.
FR-002	Functional	The system shall allow student teams to submit multimedia project artifacts for their showcase profiles.	High	Pass: A supported multimedia artifact is uploaded and associated with the correct project. Fail: An unsupported artifact is accepted.	Provides project material for the showcase.
FR-003	Functional	The system shall allow panel judges to evaluate submitted projects using a digital grading rubric.	High	Pass: A judge can enter scores for all required rubric criteria and submit the evaluation. Fail: An incomplete evaluation is accepted.	Supports consistent digital project evaluation.
FR-004	Functional	The system shall allow eligible public participants to vote for showcased projects.	High	Pass: An eligible participant can cast a valid vote and it is recorded. Fail: A voting attempt violating the rules is recorded.	Enables public participation in the showcase.
FR-005	Functional	The system shall calculate and display the project leaderboard using the applicable project scores and public votes.	High	Pass: The leaderboard reflects newly recorded valid votes and judge scores. Fail: The ranking is inconsistent with recorded results.	Provides project rankings based on results.
NFR-001	Nonfunctional	The public voting and leaderboard calculations shall update in real-time with an API response latency under 300 ms.	High	Pass: Benchmarking under simulated peak load confirms latency below 300 ms. Fail: The target is exceeded.	Ensures responsive voting and leaderboard updates.
NFR-002	Nonfunctional	The system shall prevent fraudulent or repeated public votes that violate the defined voting rules through anti-Sybil protection.	High	Pass: A repeated or unauthorized voting attempt from the same verified voting identity is detected and rejected. Fail: Multiple unauthorized votes are recorded.	Protects voting results from manipulation.
NFR-003	Nonfunctional	The system shall remain available throughout the configured project showcase and public voting period.	Medium	Pass: Monitoring confirms at least 99% uptime during the configured showcase and voting period. Fail: Availability falls below the target.	Ensures access when users need the system.
NFR-004	Nonfunctional	The system shall provide a consistent and understandable interface for student teams, panel judges, and public participants.	Medium	Pass: At least 90% of representative users complete their assigned primary task without assistance. Fail: The target is not met.	Improves usability for all stakeholder groups.
NFR-005	Nonfunctional	The system shall preserve successfully submitted project, voting, and judging records without data loss.	High	Pass: After successful submission/update, the stored record can be retrieved with all submitted values intact. Fail: Any required stored data is lost.	Maintains integrity and reliability of showcase results.

2. UML Use-Case Diagram

Paste your exported UML Use-Case Diagram below. Make sure the diagram shows all actors, primary use cases, associations, at least one «include» relationship, and at least one «extend» relationship.



Suggested minimum actors: Student Team, Panel Judge, Public Voter.

Suggested primary use cases: UC-01 Create Showcase Profile, UC-02 Submit Project Artifacts, UC-03 Evaluate Project, UC-04 Vote for Project, UC-05 View / Update Leaderboard.

3. Use-Case Flow Specification

Use Case: UC-04 — Vote for Project

Primary Actor: Public Voter

3.1 Preconditions

1. The project is published in the showcase.

2. The voting period is active.

3. The voter is eligible to participate.

3.2 Postconditions

1. A valid vote is recorded for the selected project.

2. The project's vote count and leaderboard are updated.

3. A confirmation is displayed to the voter.

3.3 Main Success Scenario

1. The public voter opens a showcased project.

2. The system displays the project's showcase details and voting option.

3. The voter selects the Vote option.

4. The system verifies the voter's eligibility.

5. The system checks the anti-Sybil voting restrictions.

6. The system records the valid vote.

7. The system updates the project's vote count.

8. The system updates the public leaderboard.

9. The system displays a vote confirmation to the voter.

10. The use case ends successfully.

3.4 Alternate Flow — Duplicate or Unauthorized Vote

4a. The system detects that the voter is not eligible or has already used the permitted vote.

4a1. The system rejects the voting attempt.

4a2. The system does not increase the project's vote count.

4a3. The system displays a message explaining that the vote cannot be recorded.

4a4. The use case ends without changing the voting result.